Based on:

1. **Negligible effect of brain MRI data preprocessing for tumor segmentation** — E. Kondrateva, P. Druzhinina, A. Dalechina, S. Zolotova, A. Golanov, B. Shirokikh, M. Belyaev, A. Kurmukov (Artificial Intelligence Research Institute, Moscow; Skolkovo Institute of Science and Technology; Moscow Gamma Knife Center, National Medical Research Center for Neurosurgery). *Biomedical Signal Processing and Control*, vol. 96, 106599, 2024; open preprint arXiv:2204.05278v4 [eess.IV], 23 Oct 2023.
2. **Scores That Hold, Benchmarks That Leak: Measuring Dataset Contamination in Public Brain-Tumor MRI Classification** — B. P. Vangala, S. Guda, L. Peddi, N. Vangala (University of Missouri; Government Degree College, Siddipet; Amar Biotech, Hyderabad). arXiv:2610.00421v1 [cs.CV], 30 Sep 2026. Preprint.

Image Preprocessing and Evaluation Integrity in Brain-Tumor MRI: The Downstream Value of Intensity Correction and the Contamination of Public Classification Benchmarks

A consolidated review of two studies, prepared as the research foundation for a Digital Signal Processing course project

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Note on sources and originality

This document condenses and reorganizes the two studies listed above, referred to throughout as Kondrateva et al. [1] and Vangala et al. [2]. Apart from Section 8, it reports no new experiments or results. Technical content, equations, figures and numerical findings are taken from the cited source; passages reproduced word-for-word are enclosed in quotation marks, and every figure and table caption names the source it was reproduced or condensed from. References cited inside the source papers are retained and renumbered in IEEE style. Page numbers for [1] refer to the arXiv v4 preprint.

Section 8 relates the material to the compiling group's own project. Statements about that project, including the duplicate check repeated on the group's data in Section 8.2, are drawn only from the group's project report and repository, which are listed separately at the end and are not part of the source literature.

Abstract

Both studies consolidated here ask what the numbers attached to a brain-tumor MRI pipeline actually measure. Kondrateva et al. [1] address the preprocessing stage. Brain MRI collected across scanners and protocols is routinely passed through a fixed recipe of rigid and atlas registration, resampling, bias-field correction, denoising, histogram matching and skull stripping before any model sees it. In a controlled ablation with two 3-D segmentation networks (nnU-Net and UNETR) on three multi-vendor glioma cohorts, they find that the intensity steps give no significant gain and can hurt (bias-field correction lowers glioblastoma Dice from 85 to 82, $p = 0.014$), while resampling to a common geometry is essential and skull stripping is the only optional step that helps. They attribute the null effect of intensity correction to a reduced divergence between tumor and healthy-tissue histograms. Vangala et al. [2] address the evaluation stage. Auditing the public Kaggle brain-tumor corpora, they find that 28.75% of one corpus's official test split and 65.74% of another's have a near-identical twin in training, that 95.5% of traceable test images share a patient with training, and that a model reading only file headers separates tumor from no tumor at 0.959 balanced accuracy. Removing the leaked images leaves network accuracy almost unchanged (mean $\Delta = -0.009$), which they read not as reassurance but as evidence that the benchmark cannot distinguish learned anatomy from learned provenance. The paper sets out the problem the two studies share, their methods, results and stated limitations, and their relevance to the compiling group's MRI enhancement and tumor-classification project.

Keywords: brain MRI; preprocessing; bias-field correction; denoising; histogram matching; skull stripping; tumor segmentation; data leakage; dataset contamination; normalized cross-correlation; shortcut learning; benchmark integrity

1. Introduction

Brain-tumor analysis from MRI is a two-ended problem. At the input end, MRI intensities are not standardized: they depend on the scanner, the protocol and the patient, so the same tissue can take very different values in different im-

ages. At the output end, a model's quality is summarized by a number measured on a test set, and that number is only as meaningful as the independence of the test set from the training data. The two studies reviewed here examine these two ends of the same chain.

Kondrateva et al. describe the input problem as one of heterogeneity across “device manufacturers, scanning protocols, and inter-subject variability”, conventionally treated with “anatomy alignment, voxel resampling, signal intensity equalization, image denoising, and localization of regions of interest”, and state that the influence of these steps “on the quality of image segmentation and on other downstream tasks in deep neural networks has never been rigorously studied” [1]. Vangala et al. describe the output problem: published classifiers on the public brain-tumor corpora report accuracies above 98%, yet evaluation on the official splits “fails to measure the integrity of the underlying dataset: whether the test partition is independent of the training partition at the image level, at the patient level, and at the acquisition-source level” [2].

The two studies therefore test two widely held assumptions: that standard preprocessing makes downstream models better, and that a high test accuracy on a public split is evidence of a working model. Section 2 summarizes the background both papers rely on, Section 3 the existing approaches and their stated limitations, and Section 4 the central research problem. Sections 5 and 6 describe the two methods, Section 7 their results, Section 8 their relevance to the compiling group's project, and Section 9 the conclusions and future directions reported by the authors.

2. Background and Related Work

2.1 MRI intensity heterogeneity and the standard preprocessing recipe

According to [1], the preprocessing recipe for brain MRI “has remained relatively stable and has been reproduced across the majority of papers on the topic from the early 2010s until now”, even as the models it feeds moved from random forests on radiomics features to fully convolutional networks, U-Nets and vision transformers. The recipe falls into four groups: intra-subject rigid alignment of the MR sequences of one patient; inter-subject alignment, either by resampling to a common voxel grid or by non-rigid registration to an atlas such as SRI24 [3]; intensity correction, comprising bias-field correction with N4 [4], image-wise z-scoring, denoising such as SUSAN [5] and histogram standardization by the method of Nyúl et al. [6]; and skull stripping, for example with HD-BET [7], to localize the brain and reduce false positives. The public BraTS benchmark [8], [9] applies reorientation, atlas registration, bias-field correction and skull stripping.

Seen as signal processing, the three intensity steps act on different parts of the image's spectrum. The bias field is “a smooth, non-heterogeneous and low-frequency signal which corrupts MRI images” [1], a slowly varying multiplicative gain that N4 estimates and divides out; SUSAN is a nonlinear smoother that averages only over neighbours of similar intensity, removing high-frequency noise while keeping edges; and histogram matching is a monotone, piecewise-linear remapping of grey levels to a reference histogram learned from training images.

2.2 Public brain-tumor classification corpora

The public classification work reviewed by [2] rests on a small family of Kaggle corpora derived in large part from the Figshare release of Cheng et al. [10], [11], which contains 3,064 contrast-enhanced T1 slices from 233 patients. The Nickparvar corpus [12] (7,023 images, four classes: glioma, meningioma, pituitary, no tumor) was assembled from Figshare, the SARTAJ corpus of Bhuvaji et al. [13] (3,264 images, the same four classes) and Br35H [14]; the Navoneel corpus of Chakrabarty [15] (253 images, tumor or no tumor) is commonly used as a second dataset. All are distributed with image-level train/test splits and no patient identifiers.

2.3 Leakage, shortcut learning and benchmark integrity

Leakage, information about the test set reaching the model during training, was formalized by Kaufman et al. [16]. In computer vision, Barz and Denzler [17] found 3.3% of the CIFAR-100 test set duplicated in training, and Recht et al. [18] built new ImageNet and CIFAR test sets to measure the resulting optimism. In medical imaging, Varoquaux and Cheplygina [19] name leakage and small test sets as main sources of over-optimism, and Roberts et al. [20] found none of more than 2,000 COVID-19 imaging models clinically usable. A related failure is shortcut learning [21]: pneu-

monia classifiers that learned hospital signatures [22] and COVID-19 classifiers that selected “shortcuts over signal” [23]. Reporting checklists such as CLAIM [24] and TRIPOD+AI [25] exist to make such problems visible.

3. Existing Approaches and Their Limitations

Table 1 brings together the existing approaches discussed in the two papers and the limitations each paper states or demonstrates. The first group concerns preprocessing and is drawn from [1]; the second concerns the evaluation of classifiers and is drawn from [2].

Table 1. Existing approaches and their limitations, as stated or demonstrated in the source papers.

Approach	What it does	Limitation reported in the sources	Source
Preprocessing before a deep segmentation model			
Full BraTS-style recipe [8], [9]	Registration, resampling, bias-field correction, skull stripping	“these steps are computationally costly and their utility for deep-learning segmentation lacks investigation”	[1]
Intensity correction: N4 [4], SUSAN [5], Nyúl matching [6]	Removes low-frequency gain, high-frequency noise, and inter-image grey-level differences	Reduces data variance, whereas augmentation, which is known to help networks [26], increases it; no consensus on which steps to apply	[1]
Earlier preprocessing studies [27], [28]	Effect of preprocessing on radiomic feature reproducibility, and on CNN segmentation	Only two close studies exist; the question had not been tested with modern networks across multi-vendor data	[1]
Evaluation of brain-tumor classifiers on public corpora			
Accuracy on official splits	Reports accuracy, often above 98%	Does not test whether the test split is independent of training at image, patient or source level	[2]
Treating SARTAJ and Nickparvar as two datasets	"External validation" across corpora	SARTAJ is physically inside Nickparvar: such a study “is, for roughly four images in five, reporting training accuracy”	[2]
Perceptual-hash deduplication	Flags near-duplicate images by a 64-bit difference hash	At Hamming radius 5 it flags 93% of a clean chest X-ray control's test split, so it cannot separate duplicates from similar anatomy	[2]
Grad-CAM [29] and SHAP explanations	Show where a network looked	Leaked and clean images draw almost the same attribution mass outside the brain (29.1% vs 28.1%), so attributions do not reveal memorization	[2]

Two observations follow. In preprocessing, the steps are justified by what they do to the image rather than by what they do to the downstream result; [1] measures the latter directly. In evaluation, a test score is taken as evidence of learning without checking that the test images are new; [2] checks this at three levels and asks whether the score would reveal the problem if it existed.

4. Problem Statement

Each paper sets out its own problem. For [1]: *given multi-protocol brain MRI, which of the standard preprocessing steps measurably change the quality of deep-learning tumor segmentation, and why?* The authors' answer is stated in their abstract: “most popular standardization steps add no value to network performance; moreover, preprocessing can hamper performance”. For [2]: *is the test partition of each public brain-tumor corpus independent of its training partition, and does contamination, where it exists, announce itself through an inflated score?* Their framing rejects the second assumption in advance: “contamination and score inflation are fundamentally distinct phenomena”, and a shortcut-laden benchmark “can hold its number steady while remaining unable to certify that any of that number was earned from anatomy” [2].

5. The Preprocessing Ablation of Kondrateva et al. [1]

5.1 Experimental pipelines

Every pipeline begins with intra-subject rigid registration of all sequences (step 1) and ends with image-wise z-scoring (step 5). Between them, one inter-subject alignment is chosen (step 2, resizing to 240 × 240 × 155 voxels; step 3, non-rigid registration to the SRI24 atlas; or step 4, resampling to 1 × 1 × 1 mm³), and after step 4 one of four optional steps may be added: 4.a bias-field correction with N4, 4.b SUSAN denoising, 4.c Nyúl histogram matching (estimated on training folds only and applied to test folds), or 4.d skull stripping with HD-BET. This gives eight pipelines (Fig. 1).

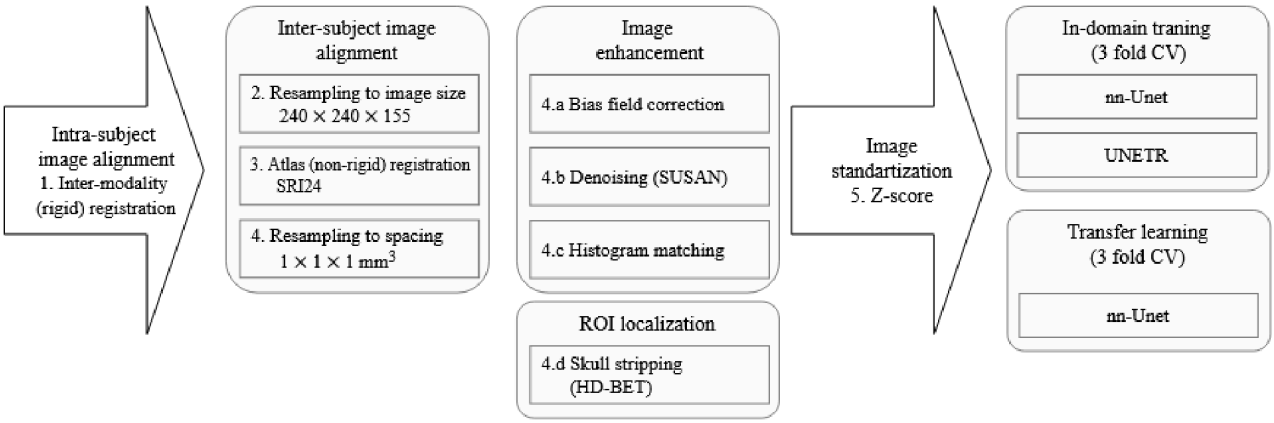


Fig. 1. "Study experimental pipeline. Steps in arrows are mandatory, steps in blocks are optional. Steps 4.a-d are performed after step 4." Reproduced from [1], Fig. 1.

Fig. 2 shows what each intensity step does to one image. The residual of N4 (the difference between the input and the corrected image) is a smooth, low-frequency field; the residual of SUSAN is fine-grained, high-frequency noise; the residual of histogram matching follows the anatomy, because the mapping changes every grey level.

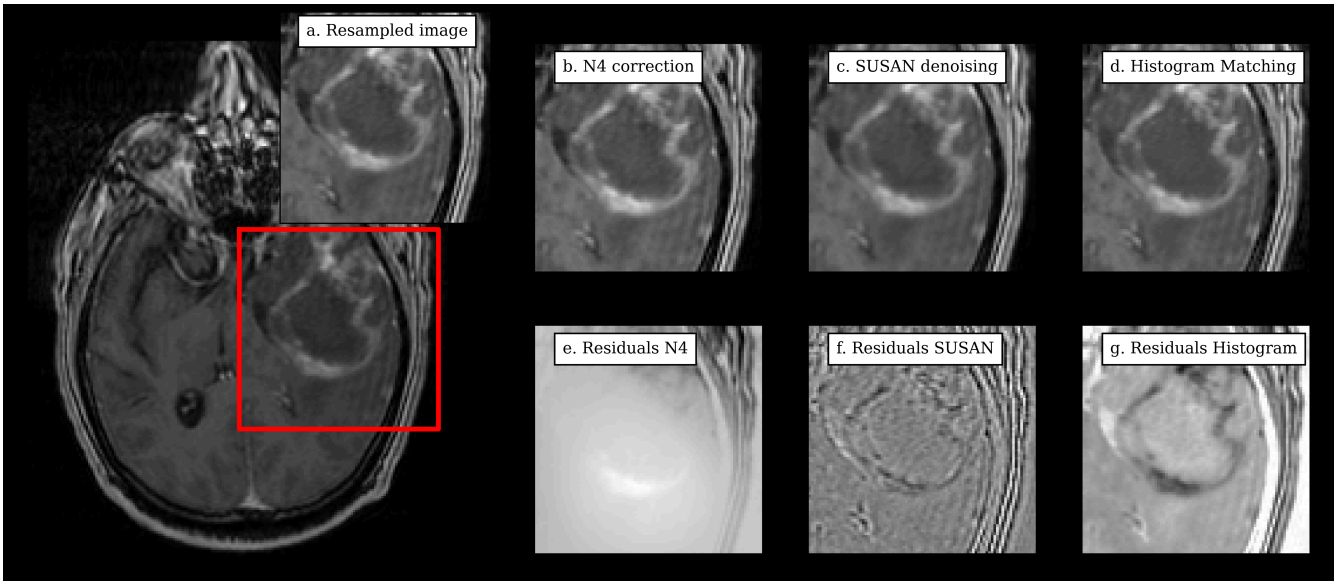


Fig. 2. The three intensity-correction methods on one contrast-enhanced T1 image with tumor: (a) resampled image, (b) N4 correction, (c) SUSAN denoising, (d) histogram standardization; (e–g) residuals between the resampled image and each corrected image. Reproduced from [1], Fig. 2.

5.2 Models and training

Two 3-D segmentation networks are used: nnU-Net [30] (NVIDIA's BraTS 2021 implementation, encoder depth 7, deep supervision, patch 128³, learning rate 3×10^{-4}) and the transformer-based UNETR [31] (embedding 512, 10 heads, learning rate 10^{-4}). Both are trained for 300 epochs or to convergence with BCE + Dice loss, with 3-fold subject-wise cross-validation, and with **no data augmentation**, "as data augmentation can interfere with measurements of preprocessing effects" [1].

5.3 Metrics

Segmentation is scored by the Dice coefficient between the predicted and reference 3-D masks A and B, and clinical usefulness by the mean absolute error of the estimated tumor volume over n subjects:

$$\mathrm{Dice}(A,B)=\frac{2\,|A\cap B|}{|A|+|B|},\qquad \mathrm{MAE}=\frac{1}{n}\sum_{i=1}^{n}\left|V^{i}_{\mathrm{true}}-V^{i}_{\mathrm{est}}\right| \tag{1}$$

To explain why a step helps or hurts, the authors compare the intensity histogram of healthy brain tissue H_{healthy} with that of tumor tissue H_{tumor} by the Kullback–Leibler divergence; a larger value means the two tissues are easier to tell apart by intensity:

$$\mathrm{KL}\left(H_{\mathrm{healthy}},H_{\mathrm{tumor}}\right)=\sum_{i=1}^{\#\mathrm{bins}}H^{i}_{\mathrm{healthy}}\,\log\frac{H^{i}_{\mathrm{healthy}}}{H^{i}_{\mathrm{tumor}}} \tag{2}$$

Differences from the reference pipeline (step 4) are tested with Welch's t-test with Bonferroni correction at α = 0.05.

5.4 Datasets

Table 2. Datasets used by Kondrateva et al. Condensed from [1], Tables 2–3.

Dataset	Subjects	Diagnosis	Labels	Vendors / protocols
TCGA-GBM [9]	102	pre-operative glioblastoma	whole tumor, enhancing tumor, tumor core	3 / ≥ 28
TCGA-LGG [9]	38	pre-operative low-grade glioma	whole tumor, enhancing tumor	3 / ≥ 11
BGPD (Burdenko) [32]	180	pre-radiotherapy glioblastoma	gross tumor volume (manual)	4 / ≥ 51

The acquisition variability is large: without resampling, voxel volumes differ by a factor of 10 in GBM (0.5 vs 5.2 mm³) and 53 in BGPD (0.1 vs 5.3 mm³) [1].

6. The Contamination Audit of Vangala et al. [2]

6.1 Three layers of contamination

For a corpus with an official training split $\mathcal{T}r$ and test split $\mathcal{T}e$, the authors define three layers. The *duplicate layer* is every test image with a pixel-level near-twin in training, including the same scan resized or re-encoded. The *patient layer* is every test image whose patient contributes another slice to training. The *source layer* is a confound rather than a set of images: when a corpus is merged from sources with different class mixes, “the acquisition source predicts the label before any anatomy is considered”. The layers are ordered by visibility: deduplication removes the first, the second survives deduplication, and the third survives both.

6.2 The duplicate criterion

Each image is reduced to a 64 × 64 grey-scale array, mean-centred and normalized to unit L_2 norm, so that the inner product of two such vectors is their zero-mean normalized cross-correlation $\rho(x,x')\in[-1,1]$. This statistic is invariant to global brightness and contrast changes and robust to JPEG re-encoding, the transformations by which one scan reappears across dataset mirrors. The duplicate-layer leak rate at threshold τ is

$$\lambda^{(1)}_{\tau}=\frac{1}{|\mathcal{T}e|}\left|\left\{x\in\mathcal{T}e:\ \max_{x'\in\mathcal{T}r}\rho(x,x')\ge\tau\right\}\right| \tag{3}$$

with the operating point τ = 0.98 and a sweep over τ ∈ {0.90, 0.95, 0.98, 0.99, 0.995}; byte-level MD5 equality is reported as a lower bound. A perceptual hash was rejected because it flags 93% of the chest X-ray control's test split. For the Nickparvar corpus the count is 460 at τ = 0.98 and 432 at 0.995, a near-flat curve that marks true duplicates rather than a threshold artefact (Fig. 3).

6.3 Patient identity recovery

Kaggle images carry no patient identifiers, but the Figshare release does. Each Kaggle image is matched back to its Figshare original by the same correlation criterion and inherits that patient key; the patient-layer leak rate is the fraction of *traceable* test images whose patient has a slice in training. It is a conditional estimate, since the SARTAJ and Br35H components carry no identifiers.

6.4 Header-only probe

To bound the non-anatomical signal, a gradient-boosted tree is trained on features containing no anatomy at all (pixel dimensions, aspect ratio, file size, bytes per pixel, image mode and JPEG quantization tables), optionally with global intensity summaries, fitted on the official training split and scored on the official test split. Its balanced accuracy is a "confound score", an "upper bound on the non-anatomical signal that provenance makes available, not the channel through which a network exploits it" [2].

6.5 Re-evaluation protocol

Nine architectures (a hand-designed CNN, ResNet-18 and ResNet-50 [33], VGG-16, EfficientNet-B0, DenseNet-121, ConvNeXt-T, ViT-B/16 and Swin-T) are trained on Nickparvar's official training split with 15% held out for validation, 224 × 224 inputs, standard augmentation, 12 epochs of AdamW, class-weighted cross-entropy and three seeds, under an equal budget. Each is scored on the official test split, on the deduplicated split, and on an external corpus with every image contained in the training corpus removed. The deduplication sensitivity is $\Delta = \mathrm{bacc}(E_{\mathrm{off}}) - \mathrm{bacc}(E_{\mathrm{dedup}})$, and calibration is tracked by the expected calibration error (ECE).

7. Results

7.1 Segmentation quality per preprocessing step [1]

Table 3. Dice (×100), mean (std), 3-fold CV. ↑/↓: significant improvement/decline against step 4 ($p < 0.05$). From [1], Table 4.

Preprocessing	nnU-Net GBM	nnU-Net BGPD	nnU-Net LGG	UNETR GBM	UNETR BGPD	UNETR LGG
1. Inter-modality registration only	44 (28) ↓	36 (29) ↓	67 (27)	39 (26) ↓	35 (30) ↓	66 (23)
2. Resampling to image size	85 (11)	73 (19)	72 (24)	82 (12)	67 (20)	66 (26)
3. Atlas registration	85 (11)	75 (16)	71 (25)	82 (13)	68 (21)	67 (25)
4. Resampling to spacing (reference)	85 (12)	74 (18)	70 (25)	83 (14)	67 (21)	67 (23)
4.a Bias-field correction	82 (13) ↓	75 (17)	67 (25)	80 (13)	72 (19) ↑	62 (22)
4.b Denoising	84 (12)	74 (17)	70 (26)	83 (13)	69 (21) ↑	65 (25)
4.c Histogram matching	83 (16)	75 (16)	68 (26)	81 (16)	68 (18)	63 (26)
4.d Skull stripping	87 (11)	76 (14) ↑	77 (21) ↑	85 (11)	72 (18) ↑	75 (19) ↑

Three findings follow. Aligning subjects to a common geometry is essential: without it Dice falls from 85 to 44 on GBM. The three alignments (steps 2–4) do not differ ($p > 0.1$), so plain resampling is as good as atlas registration. For nnU-Net the intensity steps 4.a–c give no significant improvement on any dataset ($p > 0.1$), and bias-field correction significantly lowers GBM Dice, "82 mean Dice score (step 4.a) compared to 85 mean Dice score (step 4), pvalue = 0.014". Skull stripping is the one optional step that helps, significantly on LGG and BGPD. UNETR's gains from N4 and SUSAN on BGPD are reported without an explanation; the authors suggest transformers may need more training data.

7.2 Clinical measure: tumor-volume error [1]

Table 4. Tumor-volume MAE (mL), mean (std), nnU-Net. From [1], Table 5.

Preprocessing	**GBM**	**BGPD**	**LGG**
4. Resampling to spacing (reference)	14 (13)	28 (49)	32 (31)
4.a Bias-field correction	15 (15)	27 (49)	34 (37)
4.b Denoising	13 (13)	27 (47)	32 (34)
4.c Histogram matching	14 (12)	27 (47)	42 (61)
4.d Skull stripping	10 (9) ↑	26 (42)	19 (17) ↑

Skull stripping lowers the volume error significantly on GBM and LGG, but not on BGPD (26 vs 28 mL, $p = 0.308$), which the authors take as “an argument for using clinically relevant metrics in addition to segmentation metrics” [1].

7.3 Why intensity correction does not help [1]

The authors' hypothesis is that “while these steps equalize modalities appearance across the data, they also reduce the differences between voxels' intensities within each individual image”. Table 5 supports it for bias-field correction: N4 lowers the divergence between healthy and tumor histograms on both datasets, the same step that lowered GBM Dice.

Table 5. KL divergence and Jensen–Shannon distance between healthy-brain and tumor intensity histograms, mean (std). From [1], Table A.6.

Preprocessing	**GBM, KL**	**GBM, JS**	**BGPD, KL**	**BGPD, JS**
4. Resampling to spacing	0.830 (0.450)	0.303 (0.095)	1.640 (1.015)	0.477 (0.129)
4.a Bias-field correction	0.660 (0.300)	0.259 (0.087)	1.145 (0.723)	0.465 (0.120)
4.b Denoising	0.870 (0.470)	0.308 (0.097)	1.702 (1.045)	0.484 (0.129)
4.c Histogram matching	0.770 (0.470)	0.304 (0.095)	1.650 (1.029)	0.478 (0.130)

The argument is clearest for N4: a large tumor occupies a sizeable part of the field of view, and a correction that treats part of its contrast as low-frequency gain removes some of the signal along with the bias. The authors' summary is that intensity correction “might be useful for annotators and make distinct properties of the image more recognizable for the human eye”, but that “only image alignment and voxel resampling are essential for accurate prediction with DNN models”. Preprocessing also did not help transfer learning between datasets.

7.4 Contamination inside and across corpora [2]

Table 6. Contamination of the official splits, across corpora, and at patient level. From [2], Tables I–III.

Check	**Corpus (query → reference)**	**n**	**Exact (MD5)**	**Near ($\rho \geq 0.98$)**
Test vs own training split	Nickparvar	1,600	0 (0.0%)	460 (28.75%)
	SARTAJ	394	88 (22.34%)	259 (65.74%)
	Kermany chest X-ray (control)	624	0 (0.0%)	0 (0.0%)
Corpus inside corpus	SARTAJ → Nickparvar	3,264	2,670 (81.8%)	2,705 (82.87%)
	Navoneel → Nickparvar	253	66 (26.09%)	123 (48.62%)
	Navoneel → SARTAJ	253	0 (0.0%)	98 (38.74%)
Patient level	**Corpus**	**test images**	**traceable**	**same patient in train**
Figshare patient key recovered	Nickparvar	1,600	201	192 (95.52%)
	SARTAJ	394	28	28 (100.0%)

The Figshare source holds only 233 patients across 3,064 slices, 13.2 slices per patient, a ratio that, in the authors' words, “alone makes patient-disjointness essentially impossible under a random image-level split, and every published split on this benchmark is image-level”. After every contaminated image is removed, SARTAJ keeps 559 usable im-

ages and Navoneel 130, “the only genuinely external evaluations available among these three corpora” [2]. Fig. 3 shows twins across the train/test boundary and the distribution of best-match correlations.

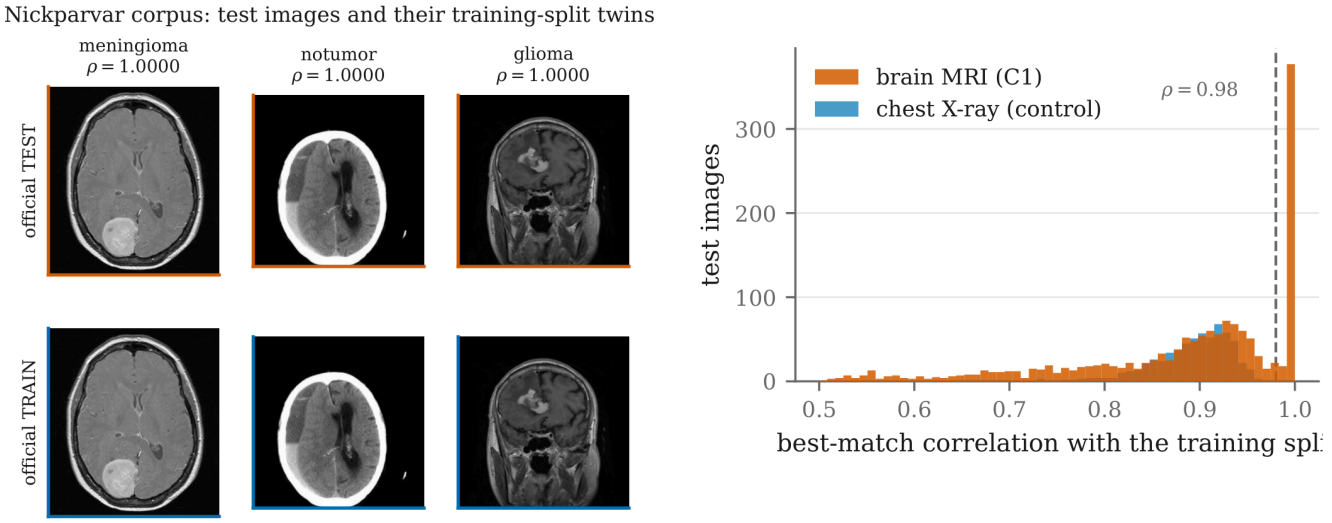


Fig. 3. Left: “Highest-correlation pairs across C1's official train/test boundary, one per class. Top row: images from the official test split. Bottom row: their matches in the official training split.” Right: “Best-match correlation against the training split, for every test image. The brain corpus has a distinct mode at ρ≈1 that the chest-radiograph control lacks entirely, despite nearly identical medians.” Reproduced from [2], Figs. 2 and 3 (C1 = Nickparvar).

7.5 A benchmark solvable from file headers [2]

Table 7. Header-only probe, balanced accuracy on the official test split. From [2], Table V.

Task	Chance	File metadata	+ intensity
Brain: tumor vs no tumor	0.50	0.959	0.966
Brain: 4-class	0.25	0.652	0.830
Chest X-ray (control)	0.50	0.496	0.499
Chest X-ray, 5-fold CV within training split	0.50	0.992	0.997

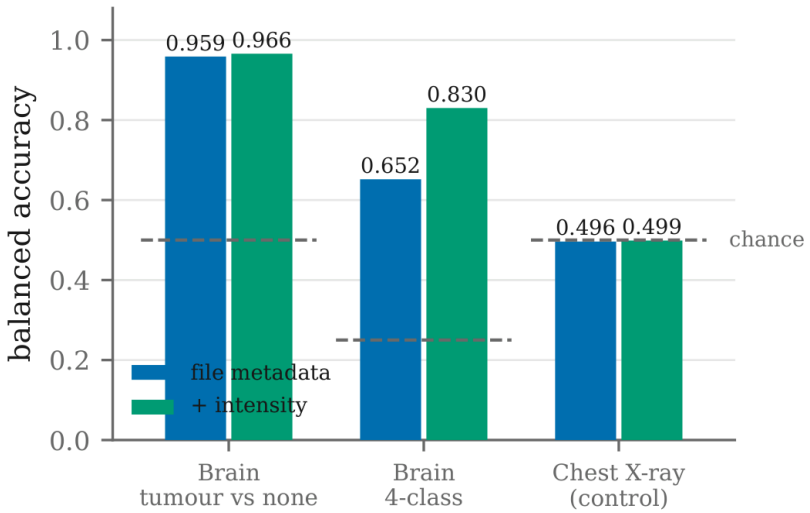


Fig. 4. “Balanced accuracy from features containing no anatomy, fitted on the official training split and scored on the official test split. Dashed lines are chance. The brain benchmark's binary task is largely solvable from file headers; the chest-radiograph control is not.” Reproduced from [2], Fig. 5.

The control row matters: header features predict the label inside the chest X-ray training split as well, but the shortcut does not transfer to its test split, whereas in the brain corpus it does. The best network reaches 0.989 on the binary task; a model reading only file size, pixel dimensions and JPEG tables reaches 0.959, at parity with fine-tuned ResNet-

18 (0.965). Hence: “A benchmark on which the headline binary task is 0.959-solvable by reading file metadata is not measuring tumor detection.”

7.6 Re-evaluation across nine architectures [2]

Table 8. Four-class balanced accuracy, mean ± s.d. over three seeds, trained on Nickparvar. From [2], Table VI (selected rows).

Architecture	Official	Deduplicated	Δ	SARTAJ (clean)	ECE in-domain	ECE external
Custom CNN	0.589±0.010	0.622±0.008	−0.033	0.491±0.038	0.078	0.185
ResNet-18	0.862±0.009	0.882±0.011	−0.020	0.924±0.010	0.057	0.081
ResNet-50	0.874±0.009	0.894±0.005	−0.020	0.830±0.021	0.053	0.069
VGG-16	0.945±0.002	0.946±0.002	−0.001	0.933±0.039	0.035	0.012
DenseNet-121	0.942±0.002	0.944±0.002	−0.002	0.893±0.011	0.036	0.022
ConvNeXt-T	0.950±0.002	0.950±0.003	−0.000	0.891±0.010	0.035	0.015
Swin-T	0.948±0.004	0.948±0.004	−0.000	0.935±0.072	0.036	0.017

Removing the 460 leaked test images changes balanced accuracy by a mean of −0.009, and by less than 0.003 for every architecture above 0.9; transfer to the cleaned external corpus costs 0.030 on average. This is the paper's unexpected result: “removing every leaked test image leaves balanced accuracy essentially unchanged: contamination of this severity does not surface as an inflated score, and deduplication repairs the number without repairing the benchmark” [2]. Even for the strongest network glioma remains the weakest class (recall 324/400; 50 gliomas called meningioma and 26 called no tumor), and the external corpus costs most in the tumor subtypes. A reverse experiment, training binary detectors on the 253-image Navoneel corpus, shows mean balanced accuracy falling from 0.745 within that corpus to 0.565 on cleaned SARTAJ; the authors conclude that such a small corpus “cannot anchor an external claim in either direction”.

7.7 What explanations reveal [2]

Grad-CAM [29] and SHAP were reduced to the fraction of attribution mass outside a brain mask taken as “the largest bright connected component after Otsu thresholding”. Leaked test images draw 29.1% (±19.4) of attribution from outside the brain and clean images 28.1% (±14.9); mean confidence is 0.923 and 0.908. Memorization is therefore not visible in the explanations. A zero-shot control intended to separate memorization from difficulty failed, because two vision–language models disagreed in sign (BiomedCLIP +0.035, CLIP ViT-B/16 −0.101), and the authors report it as such.

7.8 Discussion: where the two studies agree and where they differ

Both papers argue that a step is not justified by what it does to the image or to the number, but by a measurement that isolates its effect. Kondrateva et al. hold the network, data and training fixed and change one preprocessing step at a time, refusing augmentation so that its effect cannot mask a preprocessing effect; Vangala et al. hold the network fixed and change only the test set, and add a negative control in which the effect should be absent. Both show that a widely used step that looks correct can be neutral or harmful: intensity correction that makes images more uniform, and deduplication that makes a test split look clean. Both identify the source of the data as a confound, as scanner heterogeneity in [1] and as acquisition provenance in [2].

They differ in task and scale. [1] studies 3-D segmentation on curated, multi-sequence clinical cohorts with subject-wise cross-validation, a protocol that already avoids patient leakage; [2] studies 2-D slice classification on public Kaggle corpora where patient leakage is the central finding. [1] is peer-reviewed; [2] is a recent preprint whose own text contains small inconsistencies (for example, a 5,600/1,600 split stated for a 7,023-image corpus). Neither paper tests contrast enhancement such as CLAHE, and neither tests handcrafted features with a classical classifier, so their bearing on such pipelines is by analogy rather than direct measurement.

8. Relevance to the Compiling Group's Project

The compiling group's project, a medical image enhancement and diagnostic assistance system for brain-tumor detection from MRI, denoises each slice with median and bilateral filters, enhances it with CLAHE [34] and unsharp masking, segments a suspicious region by skull stripping and a statistical threshold, extracts 171 engineered features, detects tumors with an SVM, and types them with an SVM blended with a fine-tuned ResNet-18 [33]. It is trained and tested on the SARTAJ corpus [13], corpus C2 of [2], using the curators' split of 2,870 training and 394 test slices, and its segmentation was checked on the Navoneel corpus [15], corpus C3 of [2]. Table 9 maps each finding onto the project's components.

Table 9. Connection between the source findings and the project's components.

Source finding	Corresponding project component [P1], [P2]	Nature of the connection
Resampling to a common geometry is essential: Dice 44 → 85 on GBM [1].	Stage 2 resizes every slice to 256 × 256.	Same principle in 2-D: a fixed grid is a prerequisite for comparable features.
Skull stripping is the only optional step that helps; it reduces false positives and volume error [1].	Stage 4 strips the skull (Otsu head mask, hole filling, distance-transform peel) before thresholding.	Agreement, and stronger for a threshold-based segmenter, whose false positives come from the bright skull.
Intensity correction gives no significant gain for deep segmenters and can lower tumor/healthy divergence [1].	Stage 3 CLAHE + unsharp masking, evaluated by contrast, entropy and sharpness only.	A warning: image-quality gains (entropy 6.77 → 7.39 bits) do not show a diagnostic gain. A with/without ablation and the KL measure of Eq. (2) would test it.
Edge-preserving SUSAN denoising is neutral for nnU-Net [1].	Stage 2 bilateral filter, also edge-preserving.	Analogous filter; its value for the classical features is untested.
65.74% of SARTAJ's test split has a training twin; 88 are byte-identical [2].	The project's held-out test split (the same 394 slices).	Direct: the project's headline scores are measured on this split. Reproduced in Section 8.2.
File headers alone separate tumor from no tumor at 0.959 [2].	Detection specificity of 100% (105/105) [P1].	The perfect specificity is consistent with a provenance shortcut; every healthy test slice has a training twin.
Glioma is the weakest class even for the best network; SARTAJ's gliomas are reported as mislabelled [2].	Glioma recall 35.0% in the project's typing model [P1].	Agreement; the project also found every training glioma is 512 × 512 while the 100 test gliomas come in 64 sizes [P2].
Attribution maps do not reveal memorization [2].	Stage 7 shows every intermediate image and region measurement.	Showing evidence is necessary but not sufficient; it must be paired with a clean test set.

8.1 Preprocessing and enhancement

The project's Stages 2–4 follow the same logic as the recipe ablated in [1]: a common grid, edge-preserving denoising, an intensity step, and skull stripping. Two of the paper's findings support the project's design directly, resampling and skull stripping. The third, that intensity correction does not help a deep segmenter, cannot be transferred directly, because the project's classical branch reads intensity statistics, GLCM and LBP texture from the enhanced image, which a within-image method such as CLAHE changes in a way that a between-image method such as Nyúl matching does not. What the paper does establish is the method: the enhancement stage should be judged by an ablation on the diagnostic task, not by image-quality measures alone.

8.2 The test split

Because [2] audits the exact split on which the project's results are reported, the group repeated the duplicate check of Section 6.2 on its own copy of the data [P1]. With the same criterion, 64 × 64 zero-mean normalized cross-correlation at $\rho \geq 0.98$, it finds 88 byte-identical test files and 259 test slices with a training twin, the same counts as Table 6, stable from $\rho = 0.98$ to 0.995. Re-running the project's trained models on the two subsets gives: on the 259 twins, detection sensitivity 100% and typing accuracy 99.2%; on the 135 slices without a twin, detection sensitivity 86.7% (117/135, all 18 of the project's missed tumors) and typing accuracy 43.7%, with glioma recall 21.3% (17/80). Every healthy test slice has a twin, so specificity on unseen healthy brains could not be measured [P1].

This result differs from the paper's Table 8, where removing duplicates barely changed network accuracy. The difference is consistent with the two settings: Vangala et al. train on Nickparvar, whose test split is 29% contaminated, and

the remaining images still share patients and sources with training, whereas in SARTAJ two-thirds of the test split is contaminated, the clean third is dominated by gliomas the paper reports as mislabelled, and the project's typing model is a smaller network with a classical component. Both observations point to the same conclusion: neither the full-split nor the deduplicated number certifies tumor recognition on this benchmark, and a patient-level, source-controlled evaluation on a second dataset is required.

9. Conclusions and Future Directions

Kondrateva et al. [1] showed, for deep 3-D segmentation of multi-vendor glioma MRI, that the standard intensity-correction steps (bias-field correction, denoising and histogram matching) add no significant value and that bias-field correction can reduce accuracy, while alignment to a common voxel geometry is essential and skull stripping helps. They traced the null effect to a reduced divergence between tumor and healthy-tissue intensity distributions and proposed a minimal pipeline of alignment, resampling and z-scoring, with skull stripping optional. They name as limitations that their transfer-learning and from-scratch results “are strictly related to datasets' homogeneity and size”, that only conventional methods were tested while newer harmonization and enhancement methods “were outside the scope”, and that UNETR's response to N4 and SUSAN on one dataset is unexplained.

Vangala et al. [2] showed that the public Kaggle brain-tumor corpora are contaminated at all three levels they define: duplicated images across the official splits, test images sharing patients with training, and a source confound strong enough that file headers alone solve the binary task. They found that deduplication leaves network scores almost unchanged and concluded that “we are not claiming that published brain-tumor classifiers do not work. We are claiming that the numbers attached to them, on these corpora, are not evidence about tumor classification in the sense usually implied”. Their stated limitations are a deliberately conservative duplicate criterion that finds re-encoded scans rather than different slices of the same patient, a patient-level measure that covers only the traceable 12.6% of test images, a fixed training schedule, and a single benchmark family. As future work they propose that duplicate audits with threshold sweeps, patient-disjointness statements and header-only baselines become reporting requirements in CLAIM [24] and TRIPOD+AI [25].

For the compiling group, the two studies define the next steps: an ablation of each enhancement step on the diagnostic task, a task-relevant enhancement measure such as Eq. (2), and an evaluation on a de-duplicated, patient-level split, preferably using the Figshare corpus [11] where patient identifiers exist.

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Project sources (compiling group; not part of the source literature)

[P1] Group 11, "Medical Image Enhancement and Diagnostic Assistance System for Brain Tumor Detection," project report, Dept. of Electronics and Communication Engineering, MNNIT Allahabad, 2026, including the test-split audit of Section 7.

[P2] Group 11, project repository: *README.md*, *SYNOPSIS.md*, *docs/learn/*, and evaluation outputs *backend/outputs/metrics_binary_kaggle4.json* and *metrics_multiclass_kaggle4.json*.